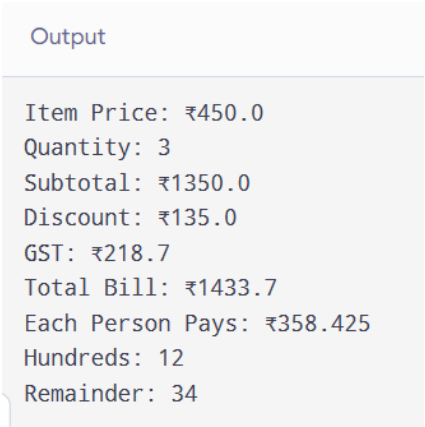
1)

```
let itemPrice = 450.0
let quantity = 3
let subtotal = itemPrice * Double(quantity)
let hasCoupon = true
let discount = hasCoupon && subtotal > 1000 ? subtotal * 0.10 : 0.0
let discountedAmount = subtotal - discount
let gst = discountedAmount * 0.18
let total = discountedAmount + gst
let sharePerPerson = total / 4.0
print("Item Price: ₹\(itemPrice)")
print("Quantity: \(quantity)")
print("Subtotal: ₹\(subtotal)")
print("Discount: ₹\(discount)")
print("GST: ₹\(gst)")
print("Total Bill: ₹\(total)")
print("Each Person Pays: ₹\(sharePerPerson)")
let amount = 1234
print("Hundreds: \(amount / 100)")
print("Remainder: \(amount % 100)")
```

Output:




2)

```
let year1 = 1900
let year2 = 2000
let year3 = 2024
let year4 = 2026
```

```
let leap1 = year1 % 4 == 0 && (year1 % 100 != 0 || year1 % 400 == 0)
let leap2 = year2 % 4 == 0 && (year2 % 100 != 0 || year2 % 400 == 0)
let leap3 = year3 % 4 == 0 && (year3 % 100 != 0 || year3 % 400 == 0)
let leap4 = year4 % 4 == 0 && (year4 % 100 != 0 || year4 % 400 == 0)
print("1900 is a leap year: \(leap1)")
print("2000 is a leap year: \(leap2)")
print("2024 is a leap year: \(leap3)")
print("2026 is a leap year: \(leap4)")
let number = 50
let isInRange = (1...100).contains(number)
print("Number is between 1 and 100: \(isInRange)")
let value = 24
let isEvenAndNotMultipleOf10 = value % 2 == 0 && value % 10 != 0
print("Even and not a multiple of 10: \(isEvenAndNotMultipleOf10)")
let nickname: String? = nil
print(nickname ?? "Guest")
```

Output:

Output

```
1900 is a leap year: false
2000 is a leap year: true
2024 is a leap year: true
2026 is a leap year: false
Number is between 1 and 100: true
Even and not a multiple of 10: true
Guest
```

3)

```
print(7 / 2)
print(7 % 2)
print(-7 % 3)
print(7.0 / 2)
print(10 - 2 * 3 + 4 / 2)
print(5 > 3 && 2 > 4 || true)
print(1 << 3)
print(6 & 3, 6 | 3, 6 ^ 3)
let x: UInt8 = 255
print(x &+ 1)
```

Output:

Output

```
3
1
-1
3.5
6
true
8
2 7 5
0
```